Tutorial 5 (Test Review)

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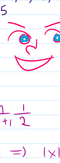
PLAN: focus on sample test 2 Q: 18,17,16,14,13,12, 11,10 8,5



18. Find the interval of convergence of

$$\sum_{n=1}^{\infty}(-1)^n\frac{x^n}{n2^n}$$

(a) $[-1,1]$ **(b)** $(-\frac{1}{2},\frac{1}{2}]$ **(c)** $[-\frac{1}{2},\frac{1}{2}]$ **(d)** $(-2,2)$ **(e)** $[-2,2]$

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1}x^{n+1}}{(n+1)2^{n+1}} \times \frac{n2^n}{(-1)^n x^n}\right| = |x|\frac{n}{n+1}\frac{1}{2}$

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}|x|\underbrace{\frac{n}{n+1}}_{\approx 1}\frac{1}{2} = \frac{|x|}{2} < 1 \Rightarrow |x|<2$

$-2<x<2$ (CANNOT BE (a),(b),(c)).

To solve this MC, only need to find if -2 converges: p-series

$a_n = \frac{(-1)^n(-2)^n}{n2^n} = \frac{(-1)^n(-1)^n 2^n}{n2^n} = \frac{(-1)^{2n}}{n} = \frac{1}{n} \to$ diverges

Hence pick (d)!

17. Which of the following series converge?

(i) $\sum_{n=1}^{\infty}\frac{4^n n^3}{(n+1)!}$ (ii) $\sum_{n=1}^{\infty}(-1)^n e^{1/n}$

(a) (i) only **(b)** (ii) only **(c)** (i) and (ii) **(d)** neither

Check (i): try ratio test:

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{4^{n+1}(n+1)^3}{(n+2)!} \times \frac{(n+1)!}{4^n n^3}\right| = 4\frac{(n+1)^3}{n^3}\frac{(n+1)!}{(n+2)(n+1)!} = 4\frac{(n+1)^3}{n^3(n+2)}$

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{4}{(n+2)}\frac{(n+1)^3}{n^3} = 0 < 1$ converges.

Check (ii): $\lim_{n\to\infty} e^{1/n} = 1$ so $\lim_{n\to\infty}(-1)^n e^{1/n} = \begin{cases}1 & n \text{ even} \neq 0\\ -1 & n \text{ odd}\end{cases}$

So it diverges.

ANS: (a) only!

16. Which of the following series are conditionally convergent?

(i) $\sum_{n=1}^{\infty}(-1)^n\frac{n}{n^2+1}$ (ii) $\sum_{n=1}^{\infty}(-1)^n\frac{n^2}{3^n+2}$

(a) (i) only **(b)** (ii) only **(c)** (i) and (ii) **(d)** neither

p-series $p=1$

(i) check abs. convergence first: $|a_n| = \frac{n}{n^2+1} \geq \frac{n}{n^2+n^2} = \frac{n}{2n^2} = \frac{1}{2n}$

So $|a_n|$ diverges.

Check if a_n converges by AST:

(1) decr: $f(x) = \frac{x}{x^2+1}$ $f'(x) = \frac{(x^2+1)-x(2x)}{(x^2+1)^2} = \frac{-x^2+1}{(x^2+1)^2} < 0$ when $x>0$

(2) $\lim_{n\to\infty}\frac{n}{n^2+1} = 0$ Hence a_n converges but not $|a_n|$

$\Rightarrow$ conditionanl convergence.

(ii) $|a_n| = \frac{n^2}{3^n+2} \geq \frac{n^2}{3^n+2^n} = \frac{n^2}{3^{n+1}}$ try ratio test on the bound:

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)^2}{3^{n+2}} \times \frac{3^{n+1}}{n^2}\right| = \frac{(n+1)^2}{n^2}\frac{1}{3}$ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)^2}{n^2}\frac{1}{3} = \frac{1}{3} < 1$

Hence $|a_n|$ converges $\Rightarrow$ a_n converges NOT CONDITIONAL CONVERGENCE!

ANS: (a)

14. Which of the following series converge?

(i) $\sum_{n=1}^{\infty}\frac{1+5^n}{1+6^n}$ (ii) $\sum_{n=1}^{\infty}\frac{\arctan n}{n^2}$

(a) (i) only **(b)** (ii) only **(c)** (i) and (ii) **(d)** neither

(i) $\frac{1+5^n}{1+6^n} \leq \frac{1+5^n}{6^n} = \frac{1}{6^n} + \frac{5^n}{6^n}$ ($r=1/6$, $r=5/6$; geometric series.)

sum of 2 conv. sequences also converge $\Rightarrow$ convergent.

(ii) $\frac{\arctan(n)}{n^2} \leq \frac{\pi}{2}\frac{1}{n^2}$ $p=2>1$ converges.

ANS: (c) – both!

13. Which of the following series converge?

(i) $\frac{1}{2}+\frac{1}{4}+\frac{1}{6}+\frac{1}{10}+\frac{1}{14}+\frac{1}{18}+\cdots$ (ii) $\sum_{n=1}^{\infty}\frac{2^{2n}3^{1-n}}{n}$

(a) (i) only **(b)** (ii) only **(c)** (i) and (ii) **(d)** neither

Check (i): Note: $\frac{1}{2} \to \frac{1}{+4} \to \frac{1}{6} \to \frac{1}{+4} \to \frac{1}{10}$ etc..

So, $a_n = \frac{1}{(4n-2)}$ But $4n-2 < 4n$

$\frac{1}{4n-2} > \frac{1}{4n}$ $\to$ diverges.

So (i) diverges.

Check (ii): $2^{2n}3^{1-n} = 2^{2n}3^{-(n-1)} = \frac{2^{2n}}{3^{n-1}}$ try ratio test:

$\lim_{n\to\infty}\left|\frac{2^{2n+2}}{3^n}\frac{3^{n-1}}{2^{2n}}\right| = \lim_{n\to\infty}\left|\frac{4}{3}\right| = \frac{4}{3} > 1$ diverges!

ANS: (d) – neither.

12. Find the values of x for which the series converges

$$\sum_{n=1}^{\infty}\frac{(2x+5)^n}{3^n}$$

(a) $(-\frac{7}{2},\frac{3}{2})$ **(b)** $(-8,-2)$ **(c)** $(-3,3)$ **(d)** $(-\frac{7}{2},\frac{3}{2})$ **(e)** $(-1,-4)$

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(2x+5)^{n+1}}{3^{n+1}} \times \frac{3^n}{(2x+5)^n}\right| = \frac{|2x+5|}{3}$ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{|2x+5|}{3} < 1$

$\Rightarrow |2x+5|<3 \Rightarrow -3<2x+5<3 \Rightarrow \frac{-8}{2}<\frac{2x}{2}<\frac{-2}{2} \Rightarrow -4<x<-1$

ANS: (e).

11. Find the sum of the following series

$$\frac{1}{3}-\frac{1}{4}+\frac{3}{16}-\frac{9}{64}+\cdots$$

(a) $\frac{1}{21}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{4}{7}$ **(d)** 4 **(e)** $\frac{1}{12}$

also negatives not even terms

$\times\frac{3}{4}$ $\times\frac{-3}{4}$

So $a_n = \frac{1}{3}\left(\frac{3}{4}\right)^{n-1}(-1)^{n-1}$ $\sum_{n=1}^{\infty}\frac{1}{3}\left(\frac{-3}{4}\right)^{n-1}$

Here $|r| = \frac{3}{4} < 1 \Rightarrow$ converges and

$\frac{a}{1-r} = \frac{1/3}{1+3/4} = \frac{1}{3} \div \frac{7}{4} = \frac{1}{3}\times\frac{4}{7} = \frac{4}{21}$ (a)

10. A sequence a_n is defined by $a_1 = 1$ and $a_{n+1} = \frac{1}{1+a_n}$. Assuming $\{a_n\}$ is convergent, find its limit.

(a) 2 **(b)** 1 **(c)** 0 **(d)** $\frac{-1+\sqrt{5}}{2}$ **(e)** $\frac{-1-\sqrt{5}}{2}$

$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} a_n = L$ (assumption)

So, $\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty}\frac{1}{1+a_n} \Rightarrow L = \frac{1}{1+L}$

$\Rightarrow L+L^2-1 = 0 \Rightarrow L^2+L-1 = 0$

quad formula: $\frac{-b\pm\sqrt{b^2-4ac}}{2a} = \frac{-1\pm\sqrt{1-4(1)(-1)}}{2(1)}$

$= \frac{-1\pm\sqrt{5}}{2}$ either $\frac{-1+\sqrt{5}}{2}$ (>0) or $\frac{-1-\sqrt{5}}{2}$ (<0) (d)

Note: a_n is never negative by defn. So, $\frac{-1+\sqrt{5}}{2}$

8. Evaluate the following sum:

$$\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$$

(a) divergent **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** $\frac{1}{4}$ **(e)** $\frac{1}{8}$

Try: partial fraction decomposition. $\frac{1}{n(n+1)} = \frac{A}{n}+\frac{B}{(n+1)}$

$\Rightarrow 1 = A(n+1)+Bn \Rightarrow 1 = n(A+B)+A \Rightarrow A=1$ $B=-1$

So $\sum_{n=4}^{\infty}\frac{1}{n}-\frac{1}{(n+1)} = \left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{6}\right)+\left(\frac{1}{6}-\frac{1}{7}\right)+\ldots$

$= \frac{1}{4}$ Ans: (d)

5. If we use the partial sum s_{10} to approximate the sum of the series

$$\sum_{n=1}^{\infty}\frac{1}{n^3}$$

estimate the error involved in the approximation.

(a) $\frac{1}{200}$ **(b)** $\frac{1}{500}$ **(c)** $\frac{1}{1000}$ **(d)** $\frac{1}{1500}$ **(e)** $\frac{1}{10^{11}}$

Thm from textbook: Remainder Estimate for the Integral Test

if $f(x)$ is a cts, positive, decr function for $x \geq 1$ and $\sum f(n)$ is convergent then $s-s_n \leq \int_n^{\infty} f(x)\,dx$

Here, $f(x) = \frac{1}{x^4}$ is cts, positive, and decr $\left(a, \frac{1}{x^4} > \frac{1}{(x+1)^4}\right)$

So, $\int_n^{\infty} x^{-4}\,dx = \frac{x^{-3}}{-3}+c\Big|_n^{\infty} = \lim_{x\to\infty}\frac{-1}{3x^3} + \frac{1}{3n^3}$

If we use s_{10} $n=10$ and then $\frac{1}{3\cdot 10^3} = \frac{1}{3000}$ (a).

Good luck on the test!!